

JUHI DAMLEY

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Claremont McKenna College

Bachelor of Arts, Computer Science

Claremont, CA

August 2024 - May 2028

Coursework completed at Harvey Mudd College through Cross-Campus Major Program

Activities: Girls Who Code (President), HMC Grutor (Grader and Tutor) for CS5, Board of Trustees Advancement

Committee, The Feedbacker (Editor), p-AI (Software Engineer), Sagehen Capital Management, Caring Key, Museum of Art Club, Women in Math

EXPERIENCE

Financial Economics Institute | Research Analyst

January 2026 - Present

- Implementing Python portfolio-optimization algorithms for the "Optimized Indexation" project, including cost- and constraint-aware routines scalable to large asset universes
- Building a robust backtesting pipeline (historical data ingestion and vectorized simulations) to evaluate tracking error, turnover, and transaction-cost impacts vs. benchmarks
- Conducting sensitivity and stress-testing to assess model robustness across market regimes
- Utilizing HPC resources to process and analyze large datasets, optimizing code for performance and scalability

The Gould Center for Humanistic Studies | Research Assistant

October 2024 - Present

- Developing professional website to showcase research, publications, courses, and related content using TypeScript, Vercel, Vite, Node, and React
- Delivered election research for Prof. Gaston Espinosa's "Religion, Race, Gender, and the 2024 Election" series spanning conferences, a five-book series, and upcoming podcast
- Managed marketing assets, social outreach, and task delegation for multi-speaker conference
- Analyzed datasets and shipped 70+ charts/visuals for presentations and publications
- Assessed scholarly viability of potential research projects by webscraping data and querying OpenAI API

Claremont McKenna College | Office of Advancement Intern

October 2024 - December 2025

- Strategically advised Board of Trustees on AI-integration opportunities and threats for Advancement
- Managed alumni data pipeline: integrated third-party platforms, digitized files, and standardized records
- Developed Python program to partially automate data updates, cutting processing time 50%+
- Cultivated custom Gemini model to automate classification of 4000+ alumnae

SKILLS AND INTERESTS

Programming Languages: Python, C++, Java, Racket, R, TypeScript

AI/Product: : Prompt Engineering, LLM Integration (Gemini/OpenAI), Claude Code, Figma

Tools / Platforms: Git, LaTeX, CRM, Adobe Photoshop, Canva, Excel, Word, Powerpoint, Vercel, Firebase, Node.js, Supabase

Relevant coursework: Abstract Algebra, Stochastic Operations Research, Linear Algebra, Statistics, Discrete Math, Calculus III, Intro to Game Theory, Data Structures and Program Development, Intermediate Microeconomics

Interests: Knitting and Crochet, Vinyl Records, Books, The Sims, Cats, Gaming, Urban Hiking, Ghibli, Concerts

PROJECTS

Complete

- Spotify Statistical Analysis** | GitHub Python — Analyzed 500k+ songs to hypothesize popularity drivers; won CMC Economics SP '25 Statistics Competition
- IE Employment Analysis** | GitHub R — Modeled and compared regional employment trends in the Inland Empire and Imperial County, developing "Job Opportunity Index" metric.
- Stylometry Model** | GitHub Python — Text-classification pipelines to compare Murakami novel translations.
- Prioritize** | Site TypeScript, Supabase — Priority-queue web app that helps users manage and organize tasks.

In Progress

- LectureTex** | Python — AI-powered lecture transcription pipeline that automatically generates structured, production-ready LaTeX math notes from live or recorded lectures. Powered by faster-whisper, Claude Haiku, aria2, and Google Colab cloud computing.
- WakeUp** | Swift — Accessibility-focused iOS and watchOS application designed to detect excessive daytime sleepiness via Apple Watch sensors and provide vibrations to induce wakefulness

HONORS & AWARDS

National Merit Scholar • Venture Capital Program Finalist • CMC SP '25 Statistics Competition Winner • AHS Merit Scholarship (Approx. \$85,000)